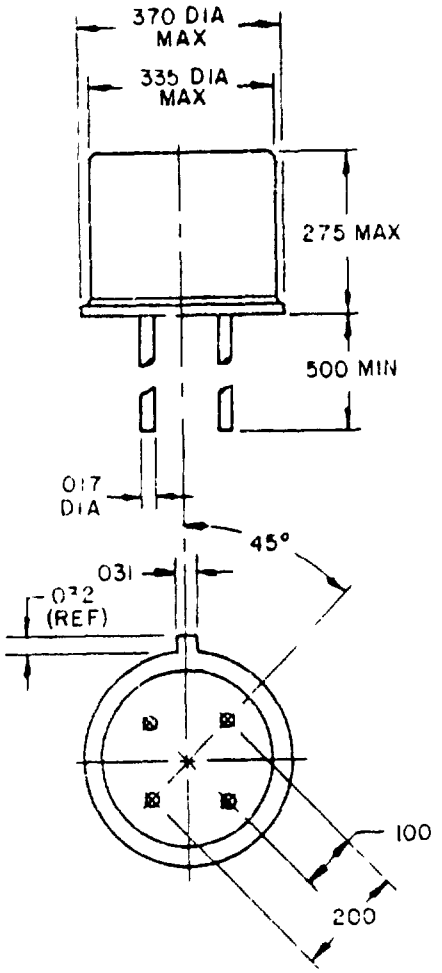
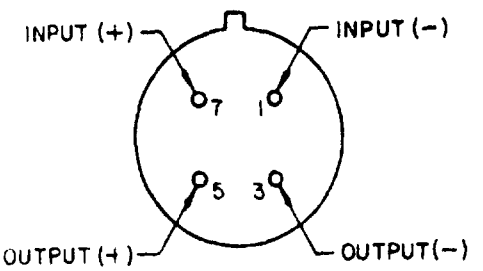
MILITARY SPECIFICATION SHEET

RELAY, SOLID STATE, HERMETICALLY SEALED, ISOLATED, 100 MA, 250 V DC, SIGNAL SWITCHING, SPST (N.O.)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-R-28750.





CIRCUIT
(BOTTOM VIEW)

INCHES	MM
.017	0.43
.031	0.79
.032	0.81
.100	2.54
.200	5.08
.275	6.99
.335	8.51
.370	9.40
.500	12.70

NOTES

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (.25 mm)
4. Terminal numbers shown above are for reference only.

FIGURE 1. Outline drawing and dimensions.

© denotes changes

FSC 5945

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REQUIREMENTS

INPUT DATA

Input voltage range 4.0 to 7 V dc. 1/

Rated turn-on voltage 5.0 V dc. 1/

Rated turn-off voltage 1.0 V dc maximum. 1/

Input current 22 mA dc maximum at rated voltage.

Turn-on time 10 microseconds maximum at rated voltage.

Turn-off time 75 microseconds maximum from rated voltage.

Bias voltage Not applicable.

OUTPUT DATA

Rated output current See figure 2.

Rated output voltage 250 V dc.

Output voltage drop .5 V dc maximum at rated voltage

Output leakage current 200 microamperes maximum at rated voltage.

Crosstalk Not applicable.

Transient voltage Not applicable.

Electric system spike Not applicable.

Overload 1.5 x rating for 5 ms, 1 percent duty cycle.

DC offset voltage Not applicable.

Waveform distortion Not applicable.

ELECTRICAL DATA

Dielectric withstanding voltage 1,000 V ac (P-P), 60 Hz.

Insulation resistance 100 megohms.

(C) Isolation 10 picofarads tested at 1 kHz.

Power dissipation 160 milliwatts maximum at rated voltage

ENVIRONMENTAL DATA

Temperature (operating and storage) -55°C to +125°C.

(C) Shock (specified pulse) MIL-STD-202, method 213, test condition F (1500 G's)

(C) Vibration MIL-STD-202, method 204, test condition H, except peak value shall be 100 G's (10 to 2,000 Hz).

Electromagnetic interference Not applicable

Resistance to soldering heat Applicable.

Salt spray (corrosion) In accordance with MIL-STD-750, method 1041

Moisture resistance Not applicable

1/ Input must be a step function Rise or fall time, as applicable, not to exceed 100 microseconds

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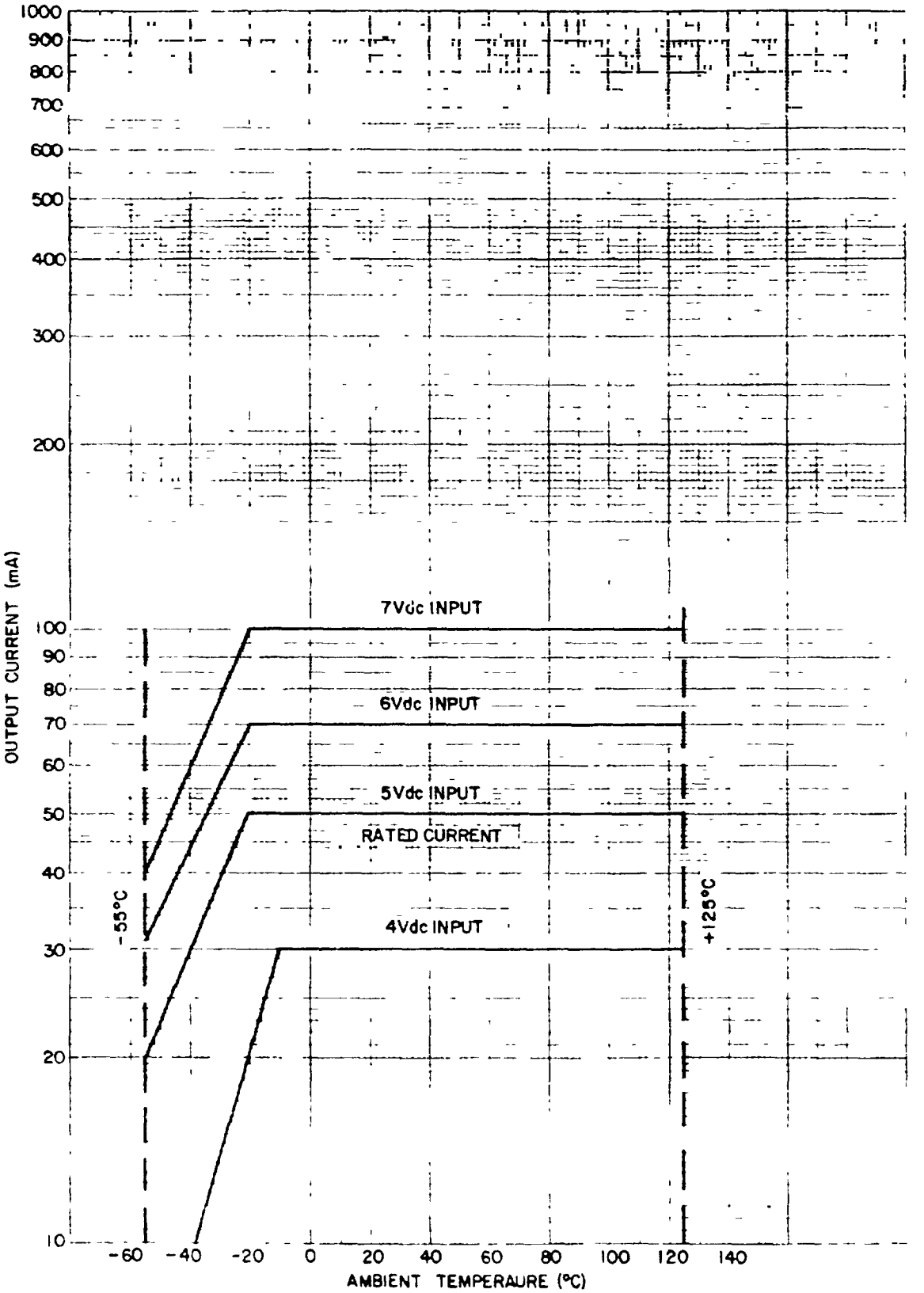


FIGURE 2. Output current vs. input control voltage and ambient temperature.

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PHYSICAL DATA·

Terminal strength (MIL-STD-202, method 211)

Pull test· Test condition A, 1 pound pull.

Bend test Test condition C, 1/2 pound load.

Twist test Test condition D.

Terminal solderability Applicable.

Dimensions and configuraton See figure 1.

Weight 5 grams (.18 ounce) typical.

Seal: Hermetic.

LIFE TEST REQUIREMENTS

Life test Applicable.

Minimum marking Military part number, functional diagram and terminal identification, date code, and manufacturer's name or source code.

Ⓒ Part number. M28750/7-001W, Y, or X to indicate the applicable screening level as specified in MIL-R-28750.

ⒸQUALIFICATION INSPECTION (reduced testing)

If the relays produced for MIL-R-28750/7 are similar in internal design, construction and material, except for the output, to the relays produced for MIL-R-28750/6, then reduced testing for qualification of MIL-R-28750/7 may be performed concurrent with or subsequent to successful qualification of MIL-R-28750/6. For reduced testing see table I.

TABLE I. Qualification inspection (reduced testing).

Inspection	Sample size
Electrical characteristics Life	6 units

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities·
Army - MI
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AR, AT, ME
Navy - AS, MC, SH
Air Force - 19

Agent
DLA - ES

Preparing activity
Navy - EC

(Project 5945-0682-3)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-R-28750/10	RELAY, SOLID STATE, HS, ISOLATED

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR
b ADDRESS (Street, City, State, ZIP Code)	☐ USER
	☐ MANUFACTURER
	☐ OTHER (Specify) _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER (Last, First, MI) – Optional	b WORK TELEPHONE NUMBER (Include Area Code) – Optional
c MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8 DATE OF SUBMISSION (YYMMDD)

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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